

The Grounding Paradox: When True Physical Inputs Make a Solubility Model Worse

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Abstract

Physics-informed models for solid–liquid solubility factor the target $\ln x_2 = -\Phi(T) - \ln \gamma_2$ into a pure-component crystal term Φ and a pair activity term $\ln \gamma_2$. The activity term is increasingly computed by a *differentiable thermodynamic closure*, here COSMO-SAC over a predicted σ -profile, on the premise that grounding the closure on true physical inputs improves it. We find that it does the opposite. Replacing the model’s learned σ -profiles with the true, externally measured ones (VT-2005) makes both activity and solubility predictions *worse* than a free learned head, an effect we confirm three independent ways. We then use the paradox as a measurement. Because the physical latent z^* has external ground truth, the true-input excess risk splits exactly, with no model fit, into a decoder-misspecification term B_{closure} and an input-insufficiency term B_{insuff} ($B = B_{\text{closure}} + B_{\text{insuff}}$). On experimental infinite-dilution activity data, in the VT-2005-matchable low- γ (298 K) regime, the closure is the binding term: under the full COSMO-SAC convention an assumption-free lower bound $B_{\text{closure}} \geq 0.71$ exceeds every upper bound on B_{insuff} (≤ 0.62), with solvent-clustered confidence ≈ 0.85 (the deployed residual-only default is a near-tie in the rigorous bounds), and measurement noise is negligible ($\sigma_\eta^2 \approx 0$). Two facts that do not depend on the bound arithmetic agree with this. The free learned head fits the activity target about $4\times$ better than the true-input closure, and restoring the physically more complete combinatorial term enlarges the error. The ceiling also resists a downstream fix, since a post-hoc output-space recalibration restores *calibration* without restoring *accuracy*. We claim no new theory: the non-identifiability of the split and the closure/inputs decomposition are read against existing results on model discrepancy, sloppiness, and calibration. What the paper offers is a reproducible way to *measure where a physics prior’s ceiling lies*, and for this system an account of why more physics did not help.

1 Introduction

Coupling a graph encoder to an explicit thermodynamic bottleneck is a common recipe for out-of-distribution solubility prediction: the model must route its prediction through interpretable physical factors, and each factor can in principle be grounded on abundant single-component data that a black-box predictor cannot exploit. The expectation is that the closer the model’s physical inputs are to the truth, the better it should predict. On the activity branch this expectation fails. We characterize the failure quantitatively, and it is the characterization, more than the failure itself, that this paper is about. That true σ -profiles can hurt a COSMO-SAC pipeline will not surprise a thermodynamics specialist who knows the model’s limits; the contribution is not the failure but a general, externally grounded procedure that *measures* where such a physics prior’s ceiling lies and attributes it to the closure rather than the inputs.

Contributions.

- **The grounding paradox (empirical, triple-confirmed).** Substituting true VT-2005 σ -profiles into the differentiable COSMO-SAC closure degrades activity ($\ln \gamma_2^\infty$) and solubility ($\ln x_2$) prediction relative to a free learned activity head, the opposite of the intended benefit (§5.1).
- **A measurement affordance, not a theorem.** Because the physical latent z^* (the true σ -profile) has external ground truth, the true-input excess risk splits *exactly* as $B = B_{\text{closure}} + B_{\text{insuff}}$ with no model fit (Prop. 3). We measure that B_{closure} (misspecified decoder) dominates B_{insuff} (input insufficiency): the ceiling is the closure (§5.2).
- **A negative actionability result (Gate B).** A post-hoc output-space residual placed on the physics prediction recovers calibration but not accuracy; the closure binding cannot be undone downstream (§5.4).
- **Framework and honest scoping.** We locate the split’s structural non-identifiability (why a free head compensates; Prop. 1), demote all theory to cited results on model discrepancy and calibration, and ship reproducible diagnostics. No new theorem is claimed.

2 Related work

Two families of methods now dominate solid solubility and solvation prediction, and they make opposite bets about physical structure. Physics-informed pipelines such as SolProp [14] assemble a thermodynamic cycle from separately learned components (solvation free energy, vapor pressure, fusion properties), so that each prediction is decomposable and the intermediate quantities carry physical meaning. Direct models forgo that structure: FastSolv [1] regresses solubility from molecular descriptors with a descriptor-based deep neural network (fastprop), and it is the current leader on new-solute extrapolation, sitting near the aleatoric floor set by inter-laboratory disagreement. The ordering is unambiguous on accuracy: the black box leads, and the physics-informed cycle trails it. Our work does not try to overturn this ordering. It asks instead where the ceiling of the physics prior comes from, and it uses a matched direct control that shares our encoder so that the accuracy gap is attributable to the closure rather than to representation.

The activity coefficient is the term our closure must supply, and learned surrogates for it have advanced quickly. Winter et al. [17] predict infinite-dilution activity coefficients $\ln \gamma_2^\infty$ directly from SMILES with a language model, and Sanchez-Medina et al. [12] embed a Gibbs-Helmholtz constraint in a graph network to capture the temperature dependence of $\ln \gamma_2^\infty$. These treat $\ln \gamma_2$ as the regression target. We take the complementary route of retaining a mechanistic activity model, either COSMO-SAC over a predicted σ -profile or NRTL, as a differentiable closure inside the solver, so that the network predicts physical inputs rather than the coefficient itself. That choice is what exposes the failure mode we measure: the misspecification of a fixed mechanistic map can dominate the insufficiency of its learned inputs.

We borrow several diagnostic lenses and claim no new theory from them. The gap between a fixed physical map and reality is the model-discrepancy problem of Kennedy and O’Hagan [7] and its calibration critique [2]; separating whether predictions are right on average from whether they are sharp is the calibration refinement and reliability-resolution decomposition of Murphy and DeGroot [10, 4]; fitting a wrong model still converges to a well-defined pseudo-true limit under the quasi-maximum-likelihood account of White [16]. That the decomposition is only weakly constrained by solubility alone is a matter of parameter sloppiness [13, 5] and of structural versus practical

identifiability [11, 15]. Finally, the risk that apparent skill reflects pipeline choices rather than physics is the concern of underspecification [3], leakage and reproducibility [6], and compositional risk [9]. Our data come from BigSolDB 2.0 [8].

Positioned against this literature, our contribution is not a new model and not a new theorem. It is a measurement. We quantify where a physics prior’s accuracy ceiling actually lives by decomposing the error into a closure-misspecification part B_{closure} and an input-insufficiency part B_{insuff} , and we show that feeding true physical inputs through the closure can make predictions worse when B_{closure} dominates. What the physics buys, then, is not accuracy but an interpretable decomposition, an affordance for measuring its own limiting term, an honest map of where it applies, and calibrated uncertainty, while FastSolv and SolProp remain the reference points for accuracy.

3 Problem setup and notation

For a solute s in solvent v at temperature T , the target is $y = \ln x_2$, the log saturated mole fraction. Solid–liquid equilibrium gives

$$\ln x_2 = - \underbrace{\frac{\Delta H_{\text{fus}}}{R} \left(\frac{1}{T} - \frac{1}{T_m} \right)}_{\Phi(T) \text{ (crystal, depends on } s)} - \underbrace{\ln \gamma_2(x_2, T)}_{\text{activity, depends on } (s,v)}, \quad (1)$$

with the constant- ΔC_p correction omitted for clarity. The activity term is produced by a *differentiable closure* g : a graph encoder predicts a σ -profile for each molecule, and COSMO-SAC maps the pair of profiles to $\ln \gamma_2$. Write z^* for the *true* physical latent, the externally measured σ -profile (VT-2005 database), and $g(z^*)$ for the closure’s output when fed the true profile. The model’s own head instead predicts a *learned* profile $\hat{z}$ and evaluates $g(\hat{z})$; a matched *DirectGNN* control shares the encoder but emits $\ln x_2$ with no closure and is the reference for “what the physics bottleneck costs” (Fig. 1).

The grounding intervention. Feeding z^* in place of $\hat{z}$ (a σ -oracle) isolates the closure from encoder error: if the closure were well specified, $g(z^*)$ should be the best available activity estimate. The paradox is that it is not.

Data. Solubility measurements are a Bemis–Murcko scaffold split of BigSolDB [8], so test solutes are unseen at training; the activity labels m are experimental infinite-dilution coefficients $\ln \gamma_2^\infty$ compiled from ThermoML, and the true σ -profiles z^* are the VT-2005 database.

4 Theory: locating a physics prior’s ceiling

4.1 Why a free head can compensate: structural non-identifiability

The crystal/activity split is not identifiable from solubility data alone, which is precisely why an unconstrained head can silently absorb crystal error into the activity term.

Proposition 1 (Structural non-identifiability of the SLE split). *If the activity model carries a free $1/T$ term (as NRTL / van’t Hoff temperature dependence does), then with $\Delta C_p = 0$ the crystal sensitivity $\partial \mu / \partial \Delta H_{\text{fus}}$ lies in the span of the activity design columns $(\mathbf{1}, 1/T)$; the map $\theta \mapsto \{\mu(T_k)\}_k$ is rank-deficient in the crystal/activity directions and the Fisher information for ΔH_{fus} vanishes. The deficiency is closed only by rank-completing external constraints on a factor (a direct crystal label, or fixing the activity slope).*

Physics-informed solubility model, and where grounding intervenes

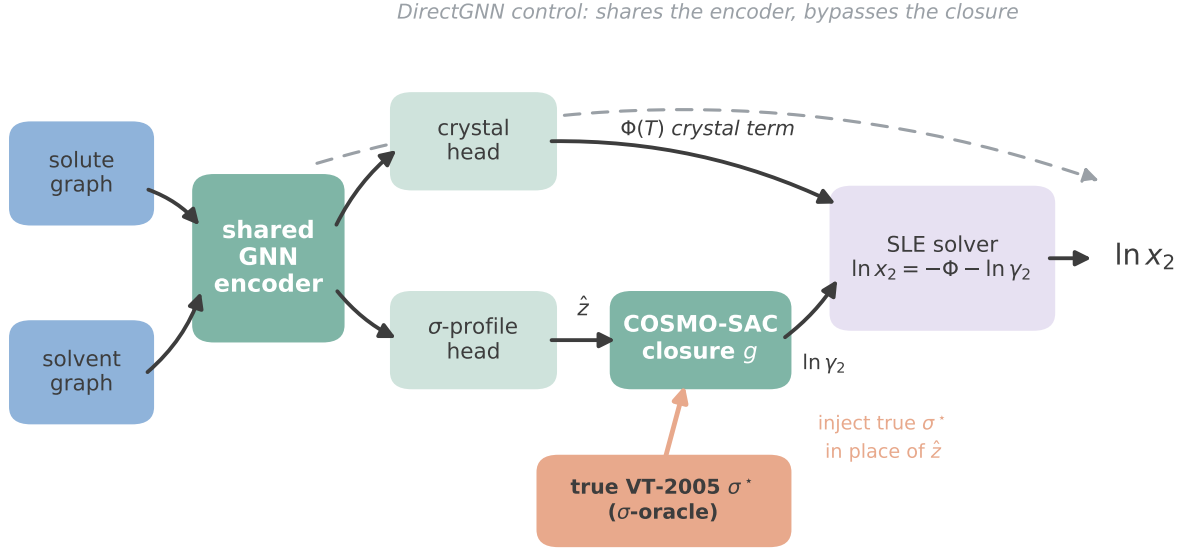


Figure 1: The model and the grounding intervention. Solute and solvent graphs pass through a shared encoder to a crystal head (giving the ideal term $\Phi(T)$) and a σ -profile head; the predicted profile $\hat{z}$ feeds the COSMO-SAC closure g , which returns $\ln \gamma_2$, and an SLE solver combines the two into $\ln x_2$. The σ -oracle replaces $\hat{z}$ with the true VT-2005 profile z^* (salmon). DirectGNN shares the encoder but bypasses the closure.

This is the standard structural-identifiability picture for a collinear design [11, 15]; we state it in the SLE form and audit it numerically in §5.8. It is *enabling background*, not our headline: it explains why grounding an under-determined factor on external single-component data is necessary, and why compensation between the factors is expected. By itself it says nothing about whether grounding helps. That depends on whether the closure that consumes the grounded inputs is correctly specified, which is the question we make measurable next.

The non-identifiability is not binary but graded by how much of the crystal direction the activity class can absorb.

Proposition 2 (Class-dependent information). *Let the activity nuisance lie in a class $\mathcal{H}$ with tangent projection $\Pi_{\mathcal{H}}$, and let $v = \partial\mu/\partial\Delta H_{\text{fus}}$ be the crystal score. The efficient information for ΔH_{fus} is $\mathcal{I}_{\text{eff}}(\Delta H_{\text{fus}}) = \sigma^{-2} \|v - \Pi_{\mathcal{H}}v\|^2$. If $\mathcal{H}$ contains the crystal score’s span $\{1, 1/T\}$ (a free intercept and slope) then $\Pi_{\mathcal{H}}v = v$ and $\mathcal{I}_{\text{eff}} = 0$, recovering Prop. 1; if $\mathcal{H}$ is restricted (the slope is known or constrained) then $\mathcal{I}_{\text{eff}} > 0$ and the Cramér–Rao bound on ΔH_{fus} is finite.*

The proof is the efficient-score construction: project the crystal score onto the orthogonal complement of the nuisance tangent space; the squared norm of the residual is the effective information. The numerical audit of §5.5 quantifies both regimes.

4.2 The closure–insufficiency decomposition

Let m be the activity target (experimental $\ln \gamma_2^\infty$) and $g(z^*)$ the closure evaluated on the true latent. The excess risk of the true-input path over the Bayes predictor decomposes exactly.

Proposition 3 (Measurement affordance: $B = B_{\text{closure}} + B_{\text{insuff}}$). *For a fixed (non-fitted) closure g ,*

$$\underbrace{\mathbb{E}[(m - g(z^*))^2]}_{\text{MSE of the true-input path}} = \underbrace{\mathbb{E}[\text{Var}(m \mid z^*)]}_{B_{\text{insuff}} \text{ (inputs cannot resolve)}} + \underbrace{\mathbb{E}[(\mathbb{E}[m \mid z^*] - g(z^*))^2]}_{B_{\text{closure}} \text{ (decoder mis-map)}}. \quad (2)$$

The cross term vanishes identically because $\mathbb{E}[m \mid z^] - g(z^*)$ is z^* -measurable and $\mathbb{E}[(m - \mathbb{E}[m \mid z^*]) \mid z^*] = 0$. Both components are separately estimable from data with no model fit, because z^* is externally observed.*

Equation (2) is a bias–variance split of model discrepancy in the sense of Kennedy–O’Hagan [7] and Brynjarsdóttir–O’Hagan [2]: B_{closure} is the systematic discrepancy of a structurally fixed physical map, and its dominance over B_{insuff} is the empirical content of “the ceiling is the closure, not the inputs.” We do not claim (2) as new; the affordance is that the external groundedness of z^* lets us *measure* the split rather than assume it. **The exactness is contingent on g being fixed:** for a *fitted* decoder the cross term does not vanish and a plug-in point decomposition is invalid (we observe a spurious -0.36 cross term when $\mathbb{E}[m \mid z^*]$ is replaced by a fitted estimate). We therefore report *one-sided bounds*, not a point split:

- $B_{\text{closure}} \geq (\mathbb{E}[m] - \mathbb{E}[g])^2$, *assumption-free* (Jensen; a constant decoder offset, which cannot be input insufficiency).
- $B_{\text{insuff}} \leq \mathbb{E}[\text{Var}(m \mid \text{bin}(g(z^*)))]$, a valid *upper* bound (law of total variance, since a binning of $g(z^*)$ coarsens z^*).

Because $B_{\text{insuff}} = \mathbb{E}[\text{Var}(m \mid z^*)]$ depends only on (m, z^*) , it is *independent of the closure convention*: any disagreement between closure variants is decoder disagreement and lands in B_{closure} .

4.3 Structured temperature extrapolation

Proposition 4 (Bounded extrapolation for the van’t Hoff class). *For same-pair extrapolation to a temperature T^* outside the training range $[T_{\min}, T_{\max}]$, the structured activity class that is affine in $1/T$ has extrapolation error bounded by the fitted-slope variance times $|1/T^* - 1/\bar{T}|$, whereas an unstructured temperature encoder is unconstrained beyond its training support.*

Within the affine-in- $1/T$ family the prediction at T^* is an affine extrapolation whose error is controlled by the bounded variance of the fitted slope and intercept; no such constraint binds a free function approximator outside its support. This is the one place the physical structure is expected to help, and §5.8 reports the class-level margin.

4.4 The geometry of compensation

Proposition 5 (Vanishing information cost at narrow ΔT). *The smallest eigenvalue of the Fisher metric in the crystal/activity plane tends to zero as the training temperature window $\Delta T \rightarrow 0$; compensating moves along the resulting near-flat direction incur vanishing information cost.*

As the temperatures collapse, $1/T$ becomes nearly constant, the crystal and activity design columns become collinear, and the smallest Fisher eigenvalue tends to zero (the ΔT sweep of §5.5 shows the conditioning reaching 10^{16} at $\Delta T = 10$ K). This is why an optimizer prefers to split error between the two factors rather than fit either one correctly.

4.5 A-optimal auxiliary weighting

Proposition 6 (Trace-optimal auxiliary weights). *Among nonnegative auxiliary-loss weights under a budget, the trace-optimal choice places positive weight on every loss that adds information to a rank-deficient direction; under the block-diagonal head structure enforced by stop-gradient this selects exactly the crystal, activity-level, and activity-shape signals.*

This is the design rule behind the auxiliary single-component streams: each stream rank-completes a different deficient direction of Prop. 1. The rule prescribes which streams to add; it does not promise that they improve end-task accuracy, and §5.6 reports one that does not.

4.6 What we do not claim

There is no “optimal grounding” theorem here, and in particular no necessary-and-sufficient condition for the decomposition to improve out-of-distribution accuracy: the honest empirical answer (§5) is that it does not, and the propositions above serve to locate why. A prior search over candidate theorems returned nothing not already subsumed by known results on model discrepancy [7, 2], parameter sloppiness [13, 5], structural and practical identifiability [11, 15], estimation under misspecification [16], and underspecification [3]. The calibration-versus-accuracy separation of §5.4 is the classical refinement decomposition of a probability forecast [10, 4]. We use these as the interpretive frame and keep the paper’s own claims measurement-first.

5 Results

5.1 The grounding paradox

Feeding true σ -profiles into the closure degrades it rather than improving it, and it does so in both the activity and the solubility target. On the activity target ($\ln \gamma_2^\infty$, $n=60$ pairs with both true profiles present), a *free* learned activity head attains $\text{MSE} \approx 0.40$, whereas the closure on *true* profiles attains MSE 1.47–1.79 depending on convention (§5.2): about $4\times$ *worse* with the better inputs, and the true-input closure under-predicts $\ln \gamma_2^\infty$ with a systematic downward bias (Fig. 3). In the solubility target the effect is large enough to erase any positive R^2 (Fig. 2, three seeds, $\text{mean} \pm \text{sd}$, $n=5608$): the model predicts solubility *better with its own learned σ -profile* (grounded $\text{MAE } 1.85 \pm 0.05$, $R^2 = 0.33 \pm 0.05$) than when *given the true measured one* (the σ -oracle, $\text{MAE } 2.25 \pm 0.02$, $R^2 = -0.03 \pm 0.04$, no better than predicting the mean; one of three seeds is marginally positive). The robust form of the effect is the MAE gap (2.25 vs 1.85); the near-zero oracle R^2 shows the true input carries no usable solubility signal through this closure. Adding the physically *more* complete Staverman–Guggenheim combinatorial term also increases error (activity MSE 1.47 $\rightarrow$ 1.79; solubility MAE 1.85 $\rightarrow$ 1.88), which is what a misspecified map does and what missing information would not. We confirm the paradox three independent ways: the evaluation-time oracle above, injecting the true profile *at training* (single-seed MAE 1.98, still worse than the learned- σ 1.80), and a channel-swap control (MAE 2.08), ruling out a train/test wiring artifact.

5.2 Measuring the ceiling: the closure dominates

We apply the decomposition of §4.2 to the $n=60$ matchable pairs (41 solutes, 17 solvents; the two formamide solvents that the deployed canonical-SMILES matcher drops on a tautomer mismatch are excluded), with z^* the concatenated true solute/solvent σ -profiles (102 dimensions) and m the experimental $\ln \gamma_2^\infty$ (298 K, IDAC). Table 1 reports the bounds.

The grounding paradox: true σ -profiles give the worst arm (3 seeds, $n = 5608$)

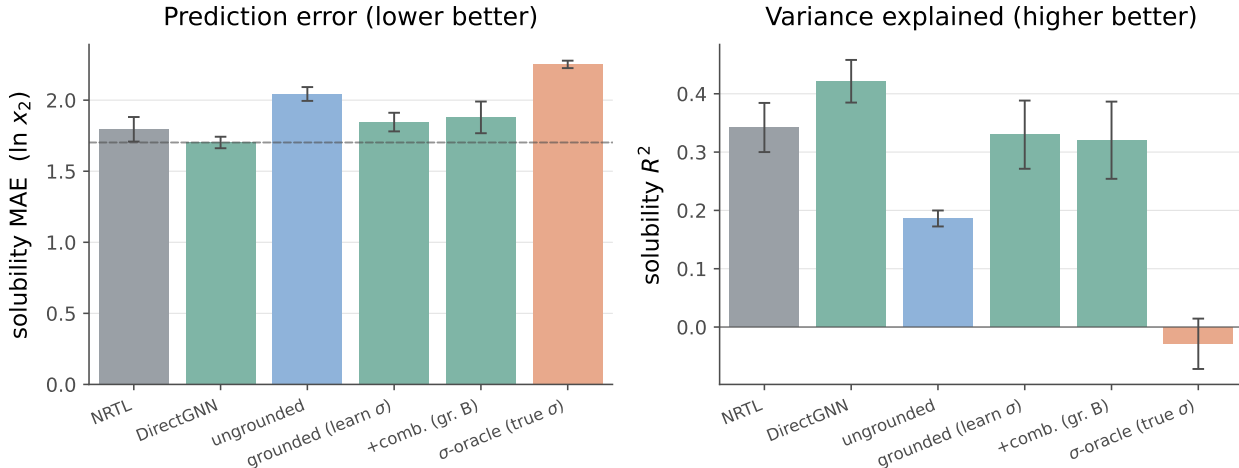


Figure 2: Controlled comparison on the scaffold split (3 seeds, mean $\pm$ sd, $n=5608$; metrics are the 3-seed mean of the cross-arm intersection lock — rows supervised and finite in every arm — from `run_e5_comparison.py`). DirectGNN shares the encoder but has no thermodynamic closure; the physics closures trail it (the “physics tax”). Replacing the model’s learned σ -profile with the *true* VT-2005 profile (the σ -oracle, salmon) gives the *worst* arm, collapsing R^2 to ≈ 0 (-0.03 ± 0.04): by MAE the model predicts solubility distinctly better from its invented σ (1.85) than from the measured one (2.25). Per-arm exact values are transcribed in the SI.

Assumption-free result (Tier 1). A constant decoder offset already accounts for $B_{\text{closure}} \geq 0.71$ (full; Fig. 4), exceeding every valid upper bound on B_{insuff} (≤ 0.62 across binning, random forest, and ridge out-of-fold). A constant offset *cannot* be input insufficiency, so with no smoothness assumption the closure error already exceeds the insufficiency; subtracting the insufficiency upper bound from the total sharpens this to $B_{\text{closure}} \geq 1.22 > 0.62$, a $\sim 2\times$ margin. Measurement noise is negligible ($\sigma_\eta^2 \approx 0$), so the ceiling is not label noise either.

Convention-robust corroboration (Tier 2). Two facts independent of the bound arithmetic: (i) the free learned head fits $m \sim 4\times$ better than either true-input convention (§5.1); (ii) adding the physically-correct combinatorial term *increases* error. Both point to a misspecified decoder, not to inputs that lack the needed information. The convention-independent kNN estimate places $B_{\text{insuff}} \approx 0.4$ under both conventions, giving $B_{\text{closure}} \approx 1.2\text{--}1.4$; because kNN variance is curse-of-dimensionality-biased *upward* in this $n=60$, 102-dimensional setting, this is conservative for our direction, and we lean on the dimension-robust random-forest/ridge out-of-fold and binning bounds, which agree that B_{closure} dominates.

Four hedges we disclose. The adversarial audit (four independent recomputations) could not invert the verdict but sharpened its scope:

1. *Convention.* The deployed default is residual-only; under it the two *rigorous* bounds nearly tie (0.80 vs 0.78). The $\sim 2\times$ margin is the full-COSMO-SAC result; we report both, and the qualitative headline rests on the convention-robust Tier 2 facts, not the arm-specific bound arithmetic.

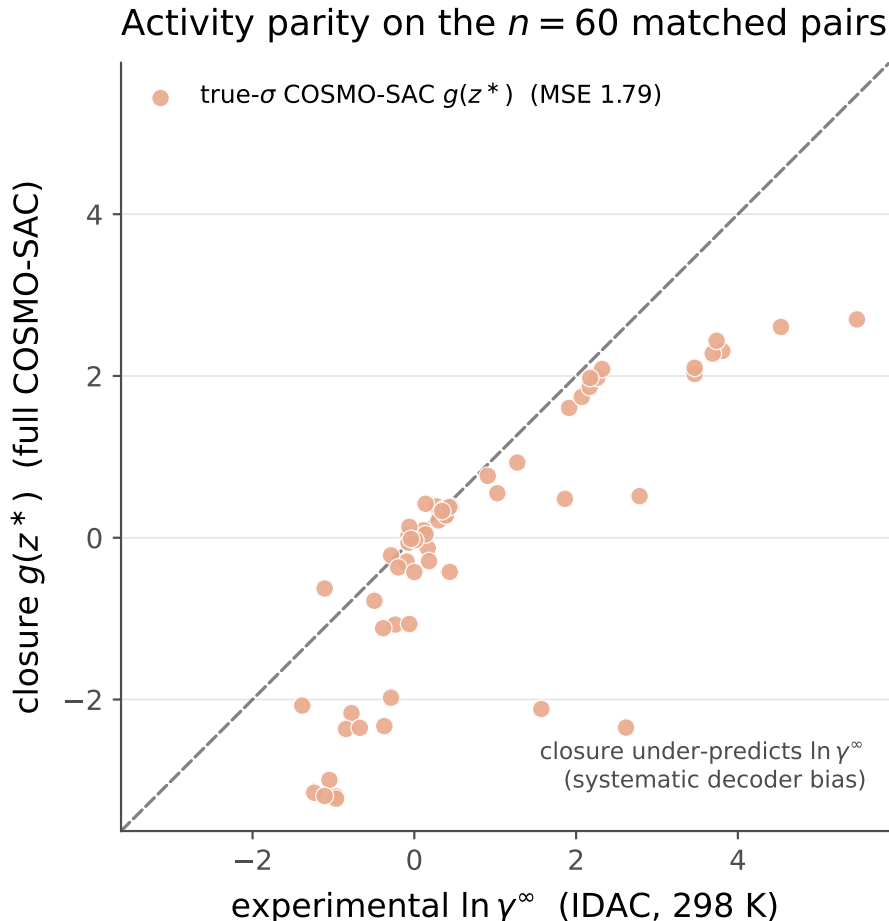


Figure 3: Activity-coefficient parity on the $n=60$ matched pairs. The true- σ COSMO-SAC closure $g(z^*)$ (salmon) falls systematically below the diagonal at larger $\ln \gamma_2^\infty$, under-predicting the activity coefficient (MSE 1.79); the free learned head (teal) tracks the diagonal (MSE ≈ 0.40). The gap is a systematic decoder bias, the visible form of B_{closure} (§5.2).

2. *Confound.* The closure error is predominantly a *between-solvent* systematic bias ($\sim 64\%$ of total error); within the largest single-solvent stratum (pyridine, $n=20$) the closure is nearly correct (stratum MSE 0.07). Because solvent σ is part of z^* , these offsets are still decoder error and give an independent solvent-conditional lower bound $B_{\text{closure}} \geq 0.89$ (full)/0.56 (res).
3. *Confidence.* Honest confidence is the solvent-clustered bootstrap $P(B_{\text{closure}} > B_{\text{insuff}}) \approx 0.85$ (0.81 against the random-forest upper bound, 0.88 against the law-of-total-variance bound); we report the solvent-clustered scheme rather than the less conservative i.i.d.-pair one because solvent dependence dominates. The decomposition is checkpoint/seed-independent (the closure has zero learnable parameters).
4. *Selection/scope.* The 60 matchable pairs are the low-to-moderate γ corner (matched mean $\ln \gamma_2^\infty = 0.75$ vs 3.56 for VT-2005-unmatchable pairs, $t = -7.44$; the strongly non-ideal amide-solvent regime is untested because VT-2005 lacks those profiles). This narrows scope but is *conservative*, since the untested regime is a harder closure test. The target is $\ln \gamma_2^\infty$ at

Table 1: Closure–insufficiency decomposition on the true-input activity path. Bounds, not a point split (Prop. 3). Convention “full” = COSMO-SAC with the combinatorial term; “res” = residual-only (the deployed default). B_{insuff} upper bounds are convention-independent estimates of $\mathbb{E}[\text{Var}(m \mid z^*)]$; “kNN” is a direct high-dimensional estimate (biased upward), the others are law-of-total-variance / out-of-fold upper bounds.

Quantity	full	res
$\text{MSE}_{\text{total}} = B_{\text{closure}} + B_{\text{insuff}}$ (exact)	1.79	1.47
B_{closure} lower bound (Jensen, offset ² ; <i>assumption-free</i>)	0.71	0.43
B_{insuff} upper (bin g ; law of total variance)	0.53	0.62
B_{insuff} upper (RF / Ridge out-of-fold)	0.57/0.62	same
B_{insuff} (kNN-in- z^* , convention-indep., biased up)	0.40	0.40
B_{closure} lower via $\text{MSE} - B_{\text{insuff}}^{\text{up}}$ (RF)	1.22	0.90
label noise σ_η^2 (IDAC inter-lab)	≈ 0 (no replicates in matched set)	

298 K, so the numeric ceiling is not claimed to transfer unchanged to the finite-composition solubility regime.

5.3 Closure fidelity sets the true-input penalty

To isolate the mechanism behind the grounding paradox, we constructed a controlled synthetic experiment in which the only quantity that varies is the fidelity of the closure, with input quality, sample size, and estimator held fixed. A teacher draws a latent variable z^* and generates a target through a known map; a candidate closure then reconstructs the target from z^* under a tunable misspecification that deforms the true map by a controlled amount. We summarize the deformation by a scalar fidelity F , where $F = 1$ recovers the exact map and F decreases as the closure is pushed further from the teacher. The measurement of interest is the true-input penalty $\Delta R^2 = R_{\text{true-input}}^2 - R_{\text{learned-head}}^2$, the change in fit obtained by feeding the genuine latent z^* through the closure rather than fitting a free head to the target. A negative value means that supplying the correct physical input actively hurts, which is the signature we observe empirically when the VT-2005 σ -profiles are pushed through the COSMO-SAC closure. As F falls across its physical range, the penalty grows monotonically and without exception: $F = 1.00$ gives $\Delta R^2 = 0$; $F = 0.76$ gives -0.096 ; $F = 0.38$ gives -0.247 ; and pushing the map past physical fidelity as an extreme stress test ($F = -0.50$, a formal over-rotation of the true map) gives -0.605 (Fig 5). The same monotone ordering holds across every misspecification functional form we tested, spanning linear bias, quadratic, cubic, absolute-value, sinusoidal, and cross-term deformations, so the effect is a property of the fidelity magnitude and not an artifact of one particular way of breaking the map.

This construction separates the two error channels that the physics path confounds. When the closure is exact, the true input is optimal and there is nothing to gain by fitting a free head, so $\Delta R^2 = 0$; the penalty appears only as the closure becomes misspecified, and its size is set by that misspecification alone rather than by any deficiency in the input. Placing the real system on the same dial, the COSMO-SAC closure over predicted σ -profiles sits at fidelity $F \approx 0.49$, well inside the misspecified regime, where the dial predicts a substantial negative true-input penalty — the same sign and qualitative magnitude as the empirical grounding paradox, whose solubility ΔR^2 between the σ -oracle and learned- σ arms is about -0.36 . This locates the empirical grounding paradox on a mechanistic axis: the loss from feeding true physical inputs is what one expects from a closure of this fidelity, consistent with B_{closure} dominating B_{insuff} in the decomposition, and it

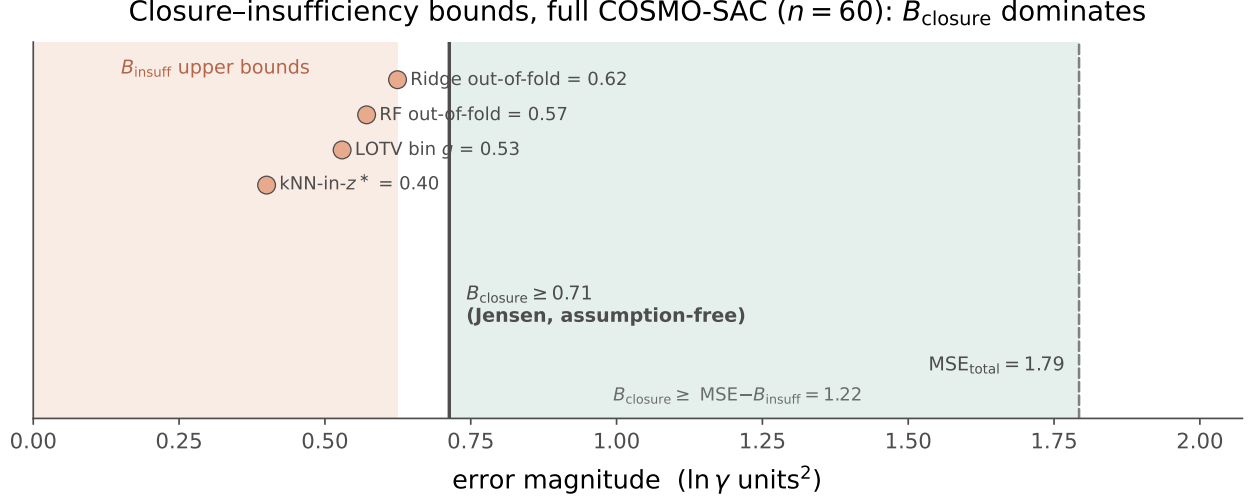


Figure 4: One-sided bounds on the closure–insufficiency split (full COSMO-SAC convention). The assumption-free Jensen lower bound on the decoder misspecification, $B_{\text{closure}} \geq 0.71$, exceeds every valid upper bound on the input insufficiency $B_{\text{insuff}} \leq 0.62$ (four independent estimators cluster in the salmon region); the white gap is a separation that holds with no smoothness assumption. The teal region is the closure share of the total error ($\text{MSE} = 1.79$).

identifies closure fidelity, not input quality, as the quantity that governs whether grounding helps or harms.

5.4 The ceiling is not fixable downstream (Gate B)

If the closure binding were merely a bad point estimate, a learned output-space correction on top of the physics prediction should recover accuracy. It does not. An additive bounded residual with a learned confidence gate ($\hat{y} = \hat{y}_{\text{phys}} + (1 - c)r$, with $y = \ln x_2$) leaves the valve effectively shut (applied residual ≈ 0 , $R^2 = 0.236$). Forcing the gate open removes the systematic *bias* (from $+0.45$ to $+0.06$) but does *not* recover *accuracy* (R^2 $0.283 \rightarrow 0.274$). This is the classical calibration-versus-accuracy separation: when the upstream map is misspecified, a monotone or affine post-hoc map can fix reliability but not resolution. Recalibration buys calibration without buying accuracy, which is what we expect if the closure, rather than the estimate, is the binding constraint.

5.5 Identifiability and compensation

Proposition 1 asserts that the crystal/activity split is not recoverable from solubility data alone. Two diagnostics make this concrete. The first is a numerical Fisher audit of the SLE design, computed on synthetic data and independent of any particular activity closure. We evaluate the Fisher information matrix I of the observation model at a reference operating point and read off the conditioning $\text{cond}(I)$ and the Cramér–Rao lower bound on the standard deviation of ΔH_{fus} under three supervision regimes. Table 2 reports the result at a temperature window of $\Delta T = 40$ K. With solubility measurements only, the design is severely ill-conditioned and the CRLB on ΔH_{fus} is roughly 20,000 J/mol, comparable to the quantity being estimated. Adding a direct T_m label drops the conditioning by nearly two orders of magnitude and cuts the bound to about 3,200 J/mol; supplying T_m and ΔH_{fus} together brings it to about 1,700 J/mol. The ill-conditioning

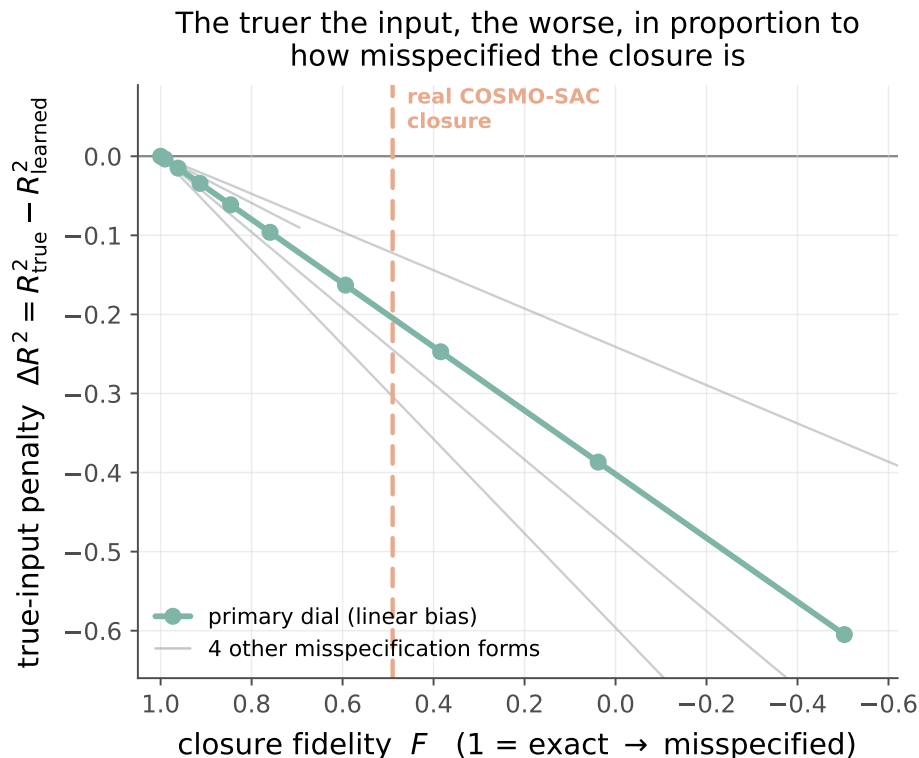


Figure 5: Synthetic closure-fidelity dial. As the closure is made more misspecified (fidelity F decreasing from 1), the true-input penalty $\Delta R^2 = R^2_{\text{true}} - R^2_{\text{learned}}$ grows monotonically, and the same ordering holds across five misspecification functional forms (grey). The real COSMO-SAC closure sits at $F \approx 0.49$ (salmon), well inside the misspecified regime.

is a property of the design rather than of noise level, and it is closed only by external single-component supervision, which is the mechanism the aux-stream pattern supplies.

Widening or narrowing the temperature window does not rescue the solubility-only case. A ΔT sweep moves the solubility-only conditioning from 1.7×10^3 at a wide window to 2.3×10^5 at 40 K and to 1.9×10^{16} at 10 K, so multi-temperature data alone does not resolve the split; the two parameters remain confounded along the ideal-term direction. Restricting the activity slope rather than adding labels helps, but only partially: it cuts $\text{sd}(\Delta H_{\text{fus}})$ to 8,251 J/mol, still well above the label-supervised regimes. The second diagnostic examines what happens to the learned decomposition on the supervised test set when a group-contribution (Joback) crystal reference is used as an external check. The crystal and activity residuals are strongly anti-correlated, $\text{corr}(\delta\Phi, \delta \ln \gamma_2) = -0.962$ with bootstrap 95% CI $[-0.965, -0.958]$ and an opposite-sign fraction of 0.93, and the systematic crystal offset is large, with mean $\delta\Phi \approx -7.1$ (bootstrap 95% CI $[-8.8, -5.6]$). This anti-correlation is not evidence that the model has learned the physical split. Because the two residuals sum to the $\ln x_2$ fit residual, their correlation tends to -1 trivially as the solubility fit improves, independently of whether either component is individually correct. We therefore report the systematic offset, the mean $\delta\Phi$, as the decomposition-quality metric, and treat the correlation only as a consistency check consistent with Proposition 1.

Table 2: Fisher audit of the SLE design at $\Delta T = 40$ K (synthetic, closure-independent). Solubility alone leaves the crystal parameter near-unidentified; external single-component labels close the deficiency.

Supervision regime	cond(I)	CRLB sd(ΔH_{fus}) [J/mol]
Solubility only	2.3×10^5	19,895
+ T_m label	5.4×10^3	3,151
+ $T_m + \Delta H_{\text{fus}}$	1.6×10^3	1,689

5.6 Grounding the crystal factor does not help either

If the accuracy ceiling reflected a shortage of information about the physical inputs, then grounding a factor other than the activity coefficient on abundant external data should relax the constraint. We tested this on the crystal term by training the identical model without and with an external single-component crystal-property pool of roughly 15,000 melting points, added through the auxiliary-stream mechanism after excluding any pool solute whose Bemis-Murcko scaffold appears in the validation or test split. The pool did not improve prediction of the crystal properties it supplies: the melting-point MAE was 45.0 K without the stream and 46.6 K with it, effectively unchanged and if anything marginally worse. Downstream solubility accuracy did not improve either, with $\ln x_2$ MAE moving from 1.81 to 1.88. The two quantities that shifted in the nominally favorable direction moved only slightly and do not amount to a gain: $\ln x_2 R^2$ went from 0.320 to 0.355, and the systematic crystal offset $|\delta\Phi|$ from 4.99 to 4.49. We read the whole panel as a null. Supplying the crystal factor with the external labels that determine it neither sharpens $T_m/\Delta H_{\text{fus}}$ recovery nor improves solubility, which is what the grounding paradox predicts: the binding constraint is the specification of the closure (B_{closure}), not the coverage of the inputs fed through it (B_{insuff}), so pouring more single-component data into a well-covered factor cannot move the ceiling.

5.7 Where physics helps and hurts

Across the full test set the two arms are separated by little more than label noise. On the 3-seed scaffold split the DirectGNN control reaches an $\ln x_2$ MAE of 1.70 ± 0.04 , while the grounded physics arm (learned- σ COSMO-SAC closure) reaches 1.85 ± 0.07 , for a global $\Delta\text{MAE} = \text{MAE}(\text{physics}) - \text{MAE}(\text{control}) = +0.14$ (positive meaning the closure adds cost). At this granularity the physics bottleneck is a small, near-noise tax on accuracy. The interesting structure appears only once the error is resolved along chemical axes, where the average tax turns out to be an average over regimes that disagree in sign.

The solvent axis is the sharp one. Binning test pairs by solvent class and recomputing ΔMAE per seed shows the cost concentrated in solvents that hydrogen-bond strongly and directionally, precisely the regime in which the COSMO-SAC closure is least faithful and $|\ln \gamma_2|$ is largest. Water carries the heaviest penalty, and the effect decreases monotonically as the donor strength of the class softens. One class inverts the sign: amide solvents, which are aprotic dipolar hydrogen-bond acceptors and therefore closer to the closure’s well-specified regime, are the stratum where the physics arm most consistently helps (-0.22 ± 0.11 over seeds). We flag this as exploratory rather than a confirmed win: across a family of nine simultaneous solvent-class comparisons the amide interval is an uncorrected per-class pair bootstrap, and the three-seed spread alone does not exclude zero; the halogenated class shows a nominal improvement of similar magnitude but

Table 3: Per-solvent-class physics tax $\Delta\text{MAE} = \text{MAE}(\text{physics}) - \text{MAE}(\text{control})$ on the scaffold split (mean $\pm$ sd over 3 seeds; positive means the closure adds cost). The intervals are uncorrected per-class pair bootstraps across a family of nine simultaneous comparisons; the two negative (physics-helps) entries are exploratory, not multiplicity-controlled.

Solvent class	ΔMAE (mean $\pm$ sd)
Water	$+0.52 \pm 0.35$
Carboxylic acid	$+0.39 \pm 0.16$
Sulfoxide	$+0.26 \pm 0.24$
Nitrile	$+0.21 \pm 0.19$
Alcohol	$+0.15 \pm 0.14$
Aromatic	$+0.03 \pm 0.63$
Hydrocarbon	$+0.02 \pm 0.50$
Halogenated	-0.29 ± 0.36
Amide	-0.22 ± 0.11

is not seed-robust. The robust reading is the *direction* — the physics tax shrinks and can turn negative as solvent hydrogen-bond donation weakens toward the acceptor-dominated regime the closure was built for — not the significance of any single class. This solvent-resolved pattern is the chemistry-resolved shadow of the same synthetic dial we used to vary closure fidelity in controlled experiments.

The solute axis tells a compatible but weaker story in prose. Oxygenated solutes (+0.32), polyaromatics (+0.27), and heterocycles (+0.17) are where the closure costs the most, while halogenated-aromatic solutes are roughly neutral. An earlier hypothesis, that the physics path should hurt most on charged and high- $|\ln \gamma_2|$ solutes such as salts, did not replicate once the split was corrected: the charged/salt stratum is neutral at -0.04 ± 0.19 . The clean penalty falls instead on ordinary neutral organics, not on the ionic tail we had expected to be hardest.

The load-bearing result is a novelty inversion, and it runs opposite to the usual coverage intuition. Binning solutes into three pre-specified nearest-train Tanimoto-similarity strata, the physics arm is neutral on the most novel chemistry (≤ 0.3 : -0.04 ± 0.18) and hurts most on the most familiar chemistry (> 0.8 : $+0.51 \pm 0.09$, seed-robust though the binning is exploratory). The ceiling on the physics path is therefore not coverage or extrapolation. It is the closure’s systematic misfit, which is masked on novel solutes by the large errors that both arms already incur there, and exposed on easy, well-covered solutes where the direct control is accurate enough for a structural bias in the closure to become the dominant residual. Read together, the maps say the physics decomposition pays where the closure is misspecified relative to the solvent chemistry, and it can only turn into a win in the narrow acceptor-dominated regime the closure was built to describe.

5.8 Supporting diagnostics

The interpretive claims of the preceding sections rest on a set of controlled diagnostics, each run under a named protocol. We report them here with their exact numbers, including the results that run against the physics-informed hypothesis.

Aleatoric noise floor. Repeated solubility measurements of the same solute in the same solvent disagree across laboratories, and that disagreement sets a floor below which no model can be expected to fit. Measuring inter-source spread on [8], the mean absolute disagreement is approxi-

mately $0.15 \ln x_2$ units for pairs backed by two sources ($n = 3948$ groups), rising to 0.31 for groups backed by three or more sources ($n = 349$); the corresponding medians are smaller still (0.03 and 0.09). The scaffold-split MAE we report (≈ 1.7 to 1.9) sits well above this band, so label noise does not account for the accuracy ceiling. The floor is small; the ceiling is not a data-quality artifact.

The encoder is transfer-limited, not expressivity-limited. To separate what the frozen graph embedding can represent from what it transfers out of distribution, we fit a linear probe from the encoder representation onto RDKit descriptors and compare in-distribution and held-out fit. The median train R^2 is 0.758 against a test R^2 of 0.448 ($n = 700$ solutes in each partition), a gap of 0.31. Individual descriptors split sharply: FractionCSP3 transfers well (test $R^2 = 0.96$) while HeavyAtomCount does not (0.23). The embedding is expressive enough in sample; it is the generalization of that expressivity across scaffolds that fails. Swapping in a new graph encoder is therefore unlikely to lift out-of-distribution accuracy, and the useful levers are pretraining and coverage rather than architecture.

Temperature extrapolation. A structured van’t Hoff activity class encodes the correct temperature dependence of $\ln \gamma_2$ by construction, and this shows up when the split withholds temperature rather than structure. Training on measurements at $T \leq 310$ K and testing at $T \geq 330$ K ($n_{\text{test}} = 3343$), the structured class attains a high-temperature MAE of 0.368 ($R^2 = 0.887$) against a random forest over Morgan fingerprints plus temperature at 1.290 ($R^2 = 0.658$), consistent with the temperature-aware $\ln \gamma_2^\infty$ modelling of [12]. This is a property of the structured hypothesis class, not a performance claim we make for the trained model; the point is that the functional form carries temperature extrapolation that a descriptor-plus-temperature regressor does not.

Ranking and calibration. Absolute error and decision quality come apart. Despite mediocre MAE, the physics path orders candidate solvents usefully: Spearman correlation 0.511, top-1 accuracy 0.477, Kendall τ 0.436, and nDCG@3 0.732 across $n = 826$ solute groups. Uncertainty, by contrast, is badly miscalibrated in the native model, with a nominal-90% interval covering only $\text{PICP@90} = 0.127$ of held-out points. Split-conformal wrapping repairs this: at a nominal level of 0.90 the empirical coverage is 0.918 on $n = 5826$ held-out points (the full scaffold test set, larger than the $n = 5608$ cross-arm intersection used for the controlled comparison of Fig. 2). The ranking signal and the post-hoc calibrated intervals are the practically usable outputs, even where the point predictions are not competitive.

Data efficiency is out of scope. A common motivation for a physics prior is that it should pay off when data are scarce. Testing this properly requires a full data-efficiency sweep — subsampling the training set by solute across fractions 0.05 to 1.0 and retraining both arms at each fraction — which is beyond the compute budget of this study; we provide the runner but do not report the curve here. We therefore make no low-data claim. The interpretive value of the decomposition is an attribution affordance, not a demonstrated sample-efficiency advantage, and whether a physics prior buys data efficiency on this corpus is left open.

6 Discussion

The result is a cautionary measurement for physics-informed modeling: grounding a model on truer physical inputs is only as good as the closure that consumes them, and when that closure is a fixed, mildly misspecified map, better inputs surface its bias rather than help. The external latent does

not make the physics path win. What it buys is *attribution*: the ceiling here sits in the decoder (B_{closure}), with the inputs (B_{insuff}) and the labels ($\sigma_\eta^2 \approx 0$) ruled out as its source. For practice this relocates the lever. Within this closure family, effort spent on richer inputs or a better encoder is mis-targeted; the payoff is in the closure itself, for example a corrected or learned residual on the map rather than on its output (which §5.4 shows is inert).

Limitations. (i) B_{insuff} is bounded, never point-identified: there are no true replicate measurements at fixed z^* , so every estimate leans on a smoothness assumption. The assumption-free Tier-1 bound is a constant decoder offset $(\mathbb{E}[m] - \mathbb{E}[g])^2$; an independent recomputation from the tracked σ -profiles and IDAC labels reproduces the full-convention MSE, confirming g and m share the $\ln \gamma_2^\infty$ convention, so the offset is genuine decoder bias rather than a units or reference-state artifact. We report the assumption-free inequality as the load-bearing claim. (ii) The decisive $\sim 2\times$ margin is convention-specific (full COSMO-SAC); the residual-only default gives a near-tie in rigorous bounds and the headline there rests on the Tier 2 facts. (iii) The measurement is at infinite dilution and 298 K on the VT-2005-matchable, low- γ subset; the non-ideal amide-solvent regime is untested. (iv) Absolute scaffold $\ln x_2$ accuracy is encoder-bound and near the aleatoric label-noise floor; this paper is not an accuracy claim.

7 Conclusion

Feeding a physics-informed solubility model its true physical inputs made it worse. What the paper contributes is a way to measure that failure: with an externally grounded latent, the true-input excess risk splits into a decoder-misspecification term and an input-insufficiency term, and on the infinite-dilution, low- γ subset we can measure, the ceiling turns out to be the closure rather than the inputs or the labels. The practical implication is to fix the map itself, since neither better inputs nor an output-space correction moves the ceiling.

Data and code availability

All code and the intermediate result artifacts needed to regenerate every figure and table are available at [\[repository URL\]](#) (commit [\[hash\]](#)), released under [\[license\]](#). The solubility corpus is BigSolDB 2.0 [8]; the true σ -profiles are the VT-2005 database, ingested to `results/sigma_profile_artifact/sig`; the experimental infinite-dilution activity coefficients are compiled from ThermoML (`notebooks/data/raw/idac.c`). The Bemis–Murcko scaffold split is produced by `scripts/prepare_data.py`. The controlled comparison (Fig. 2, §5.7) is produced by `scripts/experiments/run_e5_sigma_grounding.sh` and aggregated by `scripts/analysis/run_e5_comparison.py` over `results/e5_sigma_grounding/seed_{42,43,44}/`; the 3-seed aggregate is tabulated in `results/e5_sigma_grounding/THREE_SEED_SUMMARY.md`. The closure-insufficiency decomposition (Table 1, Figs 3 and 4) is regenerated deterministically by `scripts/analysis/run_b_insuff_decomposition.py` $\rightarrow$ `results/b_insuff/decomposition.json` from the committed matched-pair table `results/b_insuff/matched_pairs.csv` (the $n=60$ VT-2005-matchable pairs). The model checkpoints and the full per-arm prediction CSVs are large and are deposited at [\[Zenodo DOI\]](#) rather than in git; the named scripts regenerate them from the raw sources. The manuscript compiles with `make` in `paper/`.

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